Carbon Taxes and Clean Technology Transition in an International Environment

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Motivation

- Carbon taxes are generally effective in reducing domestic emissions and encourage the adoption of clean technologies (wind, solar etc)
- However, if only a subset of countries adopt carbon taxes, efforts to reduce global emissions can be undermined.
- "The reason why international climate agreements fail is the strong free-riding incentives in these agreements." Nordhaus (2015, AER).

Research Question

- What is the optimal climate policy and key trade-offs in an international environment ?
- What are the effects of international cooperation/non-cooperation on the transition to clean technology?
- Build a climate-macro model to:
  - Theoretically characterise key trade-offs in setting carbon taxes.
  - Quantitatively study the effects of climate policy on the economy, climate, and welfare.

Results Preview

- **Cooperation outcome:** The global optimal carbon taxes are $\{\hat{\tau}_{US} = 6\%; \hat{\tau}_{China} = 30\%\}$, with significant environmental benefits, more clean R&D, but at the cost of lower growth.

- **Non-cooperation** The same optimal taxes but applied unilaterally fail to deliver climate benefits and discourages clean R&D.

  Instead, production relocates to the no-tax country and global emissions increase.

- **Nash Equilibrium:** A country has an incentive to undercut the other's tax & there is no positive welfare-improving equilibrium tax.

Literature

Environmental policy in an international setting

Farrokhi & Lashkaripour (2025): Examine whether trade policies in a multi-country setting can deliver climate benefits.

This paper: Dynamics, Growth and Innovation.

Carbon taxes and technological change

Acemoglu et al. (2016): Clean and dirty technologies compete in production with R&D. Carbon taxes and research subsidies encourage production & clean innovation.

This paper: Optimal policy and transition to clean in an international setting.

Overview

Model

Results

Model Set-Up

- Two countries: Home and Foreign (denoted *).
- Homogenous final good produced by combining Home and Foreign intermediate goods.
- Household includes unskilled workers (measure one) and "scientists" (measure L^s) supplied inelastically.
- Production input is unskilled labour.

Intermediate Sector

- Continuum of intermediate sectors/goods $j \in [0, 1]$
- Each good is produced with either clean or dirty technology.
- Each sector j is a dynamic contest between two firms:
  One specialising in dirty technology research.
  And the other specialising in clean technology research.

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  $e_{j,t}$: Exhaustible resource used for dirty production.

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- Taxes on dirty costs: τ^d_t

R&D and Quality Ladder

- Productivity of clean and dirty firms in each sector is represented by a quality ladder that evolves as a result of R&D.

- Each step in the quality ladder is a proportional productivity improvement of $\lambda > 1$

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- Hire scientists for innovation.

  $s_{j,t}$: Number of scientists hired

  The probability of a successful innovation is $p_{j,t}(s_{j,t})$

R&D and Quality Ladder

- Let initial productivity $\theta_{j,0} = 1$.

  Labour productivity of a firm in sector j at time t is

$$\theta_{j,t} \;=\; \lambda^{n_{j,t}}$$

  $n_{j,t}$: Number of steps climbed since $t = 0$

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- Relative productivity of the dirty firm to the clean firm at time t is

$$\frac{\theta^d_{j,t}}{\theta^c_{j,t}} = \lambda^{n^d_{j,t} - n^c_{j,t}}$$

  $n^d_{j,t} - n^c_{j,t}$: Technology gap between the two firms.

Technology Choice

Lemma 1

The tax-adjusted technology choice threshold is

$$\text{Clean leader if} \qquad n^d_{j,t} - n^c_{j,t} \leq m_{j,t}$$

$$\text{Dirty leader if} \qquad n^d_{j,t} - n^c_{j,t} > m_{j,t}$$

where

$$m_{j,t} = \frac{1}{ln\,\lambda}\left[ln\left((1+\tau^d_t)\tilde{P}_t\right)\right]$$

is the price-adjusted policy gap in each sector.

Value Functions

$$V^d_{j,t}(n^d_{j,t} - n^c_{j,t}; m_{j,t}) = \max_{s^d_{j,t}} 1_{\{n^d_{j,t} - n^c_{j,t} > m_{j,t}\}} \pi^d_{j,t} - w^s_t s^d_{j,t}$$

$$+ \rho \mathbb{E} V^d_{j,t+1}(n^d_{j,t+1} - n^c_{j,t+1}; m_{j,t+1})$$

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Carbon Cycle and Preferences

- Total Home and Foreign carbon emissions at time t is

$$K_t = \kappa \int_0^1 e_{j,t}\, dj \qquad K_t^* = \kappa \int_0^1 e_{j,t}^*\, dj$$

κ : Units of carbon per exhaustible resource.

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- Atmospheric carbon concentration: Accumulation of emissions

$$Z_t = \sum_{l=0}^{t} (1-\delta_l)(K_{t-l} + K_{t-l}^*)$$

- Temperature change: $T(Z_t)$
- Household preferences: $U_0 = \sum_{t=0}^{\infty} \beta^t \left[u(c_t) + v(T_t)\right]$

Overview

Model

Results

Calibration

- Symmetric countries (US+EU); Asymmetric countries (US+China)

- To solve the model
  - 23 parameters
  - the price path of exhaustible resources $\{p_t^e\}_{j=0}^{\infty}$
  - and initial technology gaps distribution $\{n_{j,0}^d - n_{j,0}^c\}_{j=0}^{1}$

- Calibrated to match moments in 2015.

- One period is 10 years.

Symmetric Countries

Moment (US+EU)	Data	Model
R&D/Sales ratio 2015 (%)	2.97	2.96
Actual emissions 2015 (Gtn)	8.3	8.29
Temperature rise by 2100 (°C)	2	1.75
Annualised emissions growth rate 2005-2015 (%)	-1.55	-0.91
Annualised GDP growth rate 2005-2015 (%)	2.48	2.26

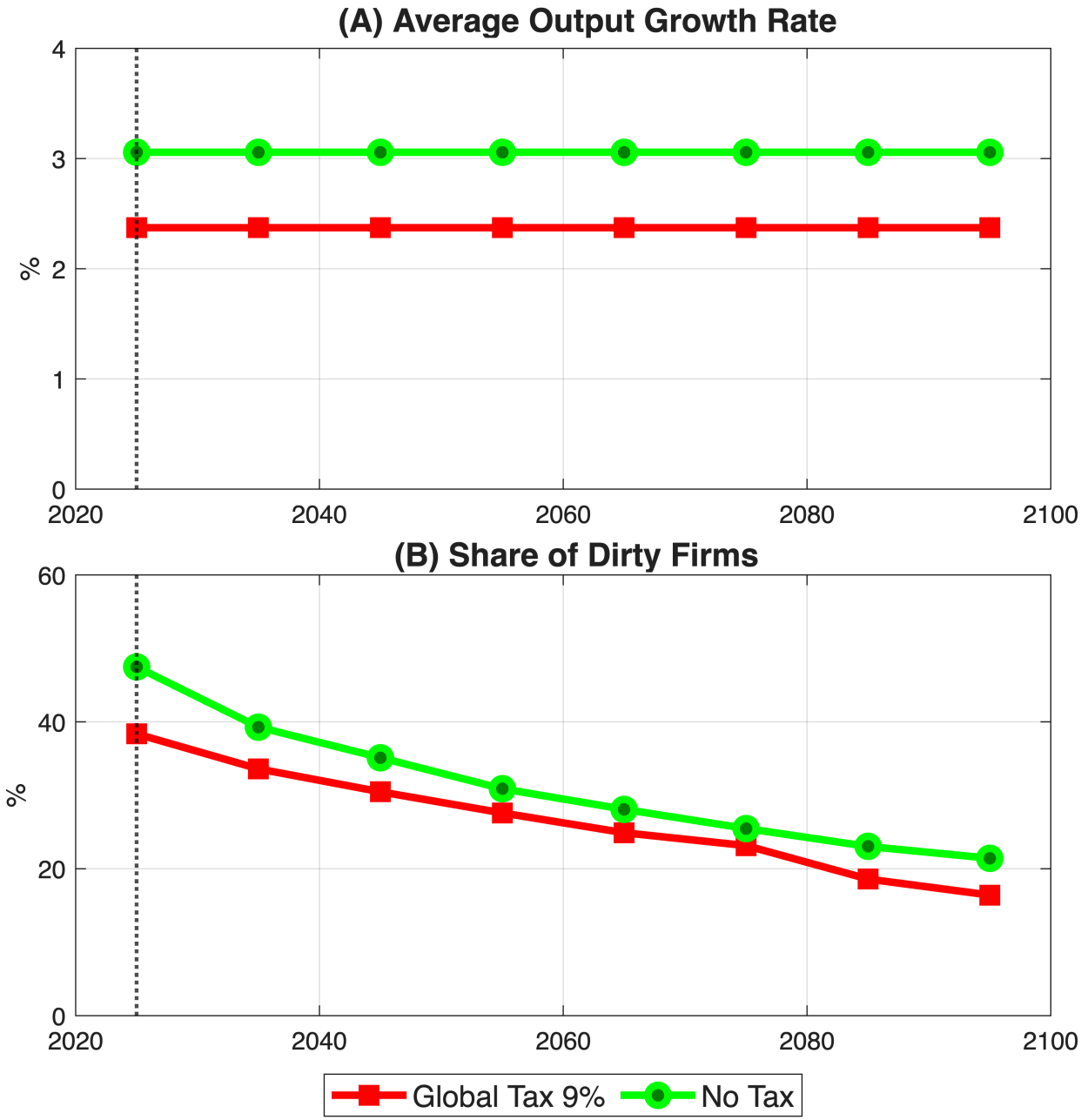
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A planner maximises HOME and FOREIGN social welfare function

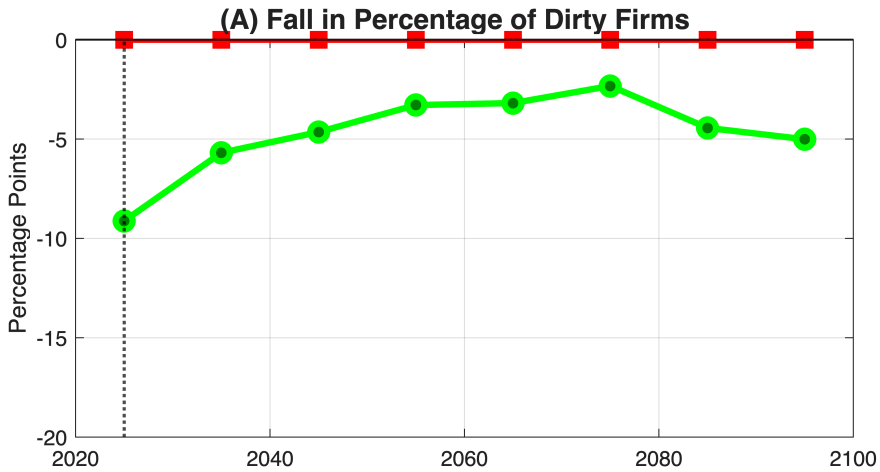
$$\hat{\tau^d}, \left(\hat{\tau^d}\right)^* = \arg \max_{\tau^d, (\tau^d)^*} \sum_{t=0}^{\infty} \beta^t \left[u(c_t) + v(T_t) + u(c_t^*) + v(T_t)\right] = 9\%$$

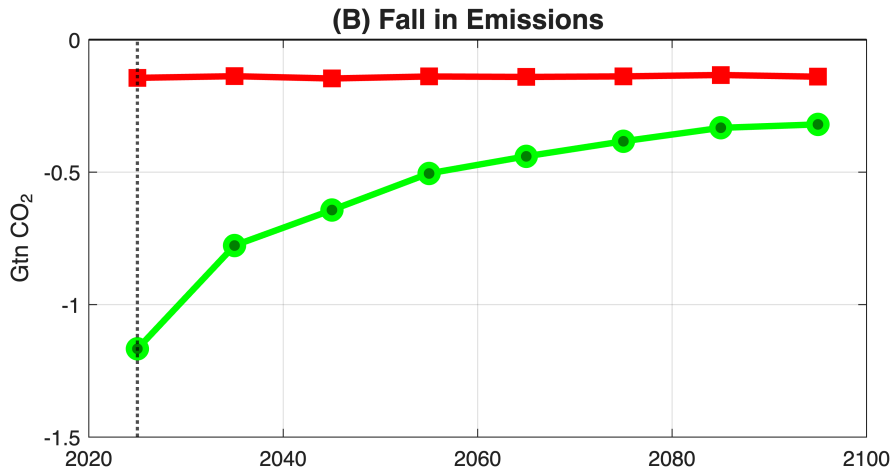
Effects of Carbon Tax



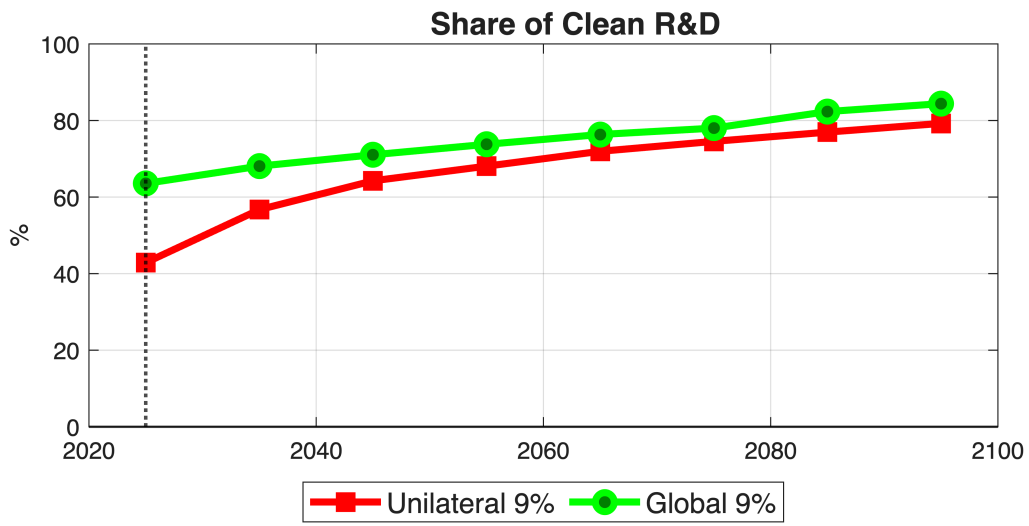
Unilateral v/s Global

Panel 1: Change in share of dirty firms and emissions relative to the no-tax baseline





Panel 2: Clean R&D intensity



Asymmetric Countries

	US		China	
Country-specific moments	Data	Model	Data	Model
Actual emissions 2015 (Gtn)	5.3	5.3	9.9	9.9
Annualised emissions growth (%)	-1.24	-1.15	5.3	5.64
R&D/Sales ratio 2015 (%)	3.9	3.88	2.02	2.01
Annualised GDP growth (%)	3.44	2.42	9.5	9.17

Global moments	Data	Model
GDP ratio (US/China) 2015	1.61	1.54
Temperature rise by 2100 (°C)	2	2.01

Asymmetric Countries

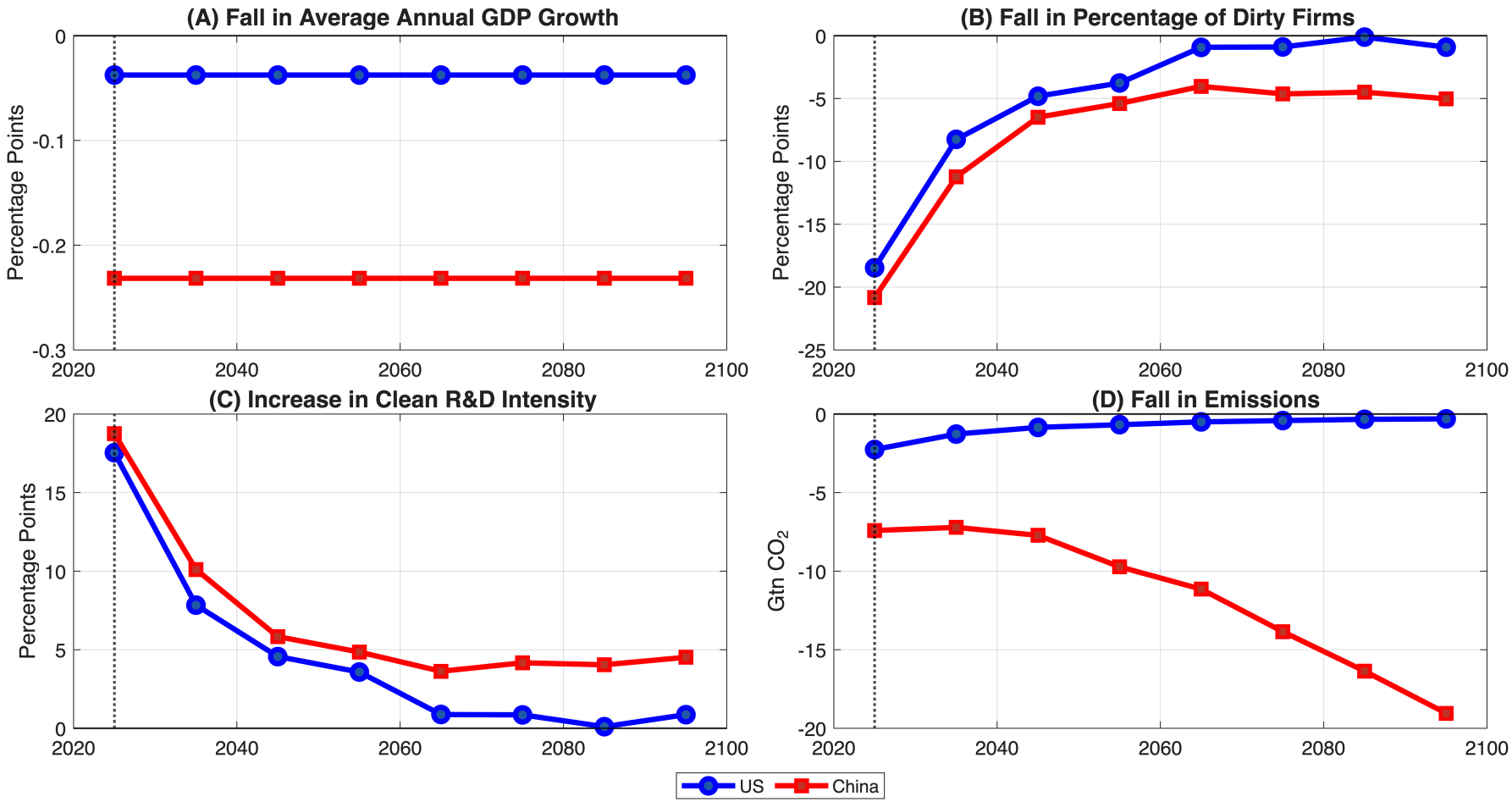
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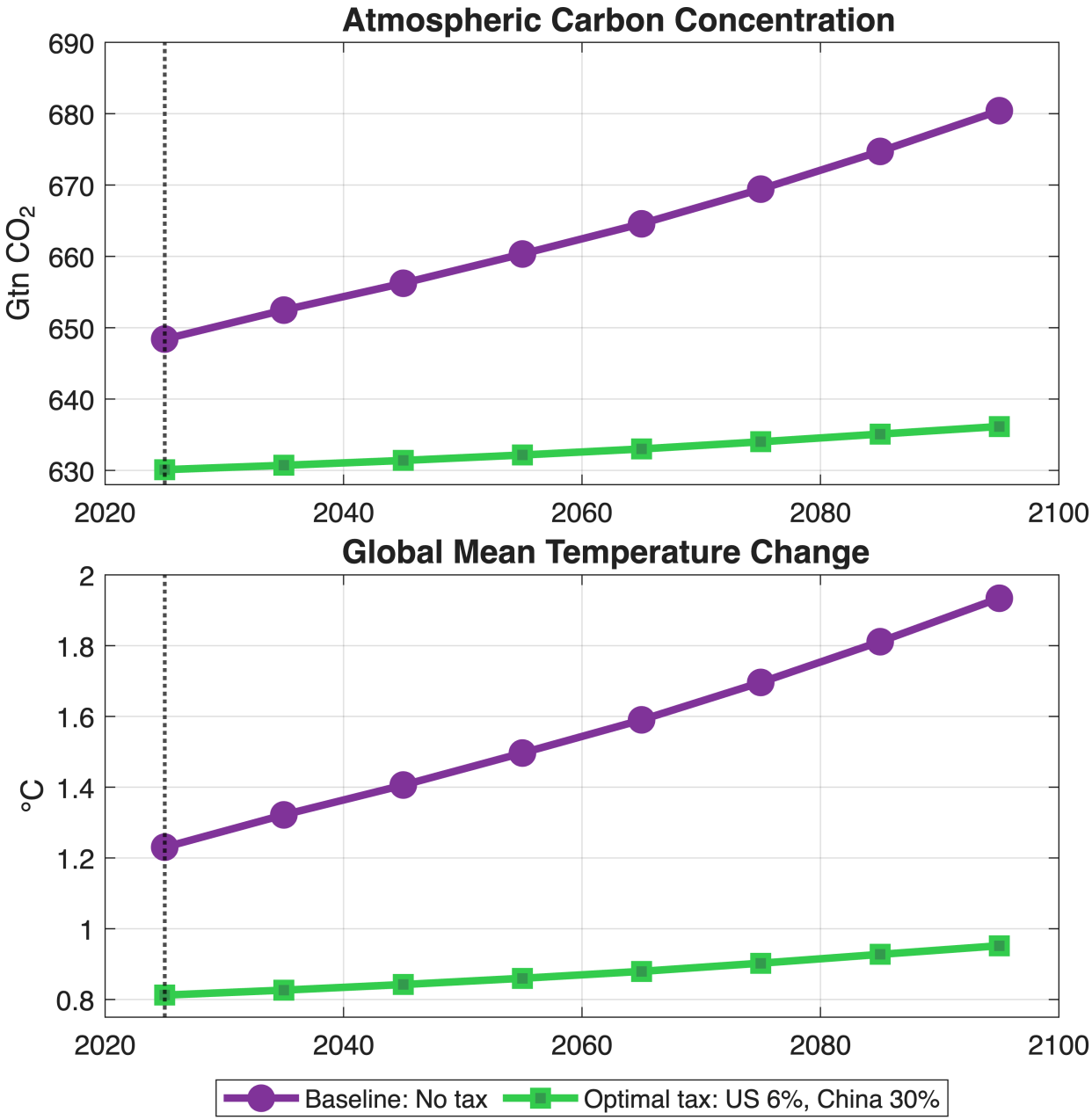
Optimal policy: $\hat{\tau^d_{US}} = 6\%$ $\hat{\tau^d_{China}} = 30\%$

Carbon taxes (Asymmetric countries)

Panel 1: Change in GDP growth rate, share of dirty firms, clean R&D intensity and emissions relative to no-tax baseline



Global environmental outcomes (Asymmetric)



Summary

- Theoretically characterise key trade-offs in setting carbon taxes internationally.

- Quantitatively study the effects of climate policies on the economy, climate and welfare.

  Key takeaway: Unless there is coordination and enforceability of climate policies, countries will have an incentive to free-ride and there will be limited climate benefits.

Thank You

Overview

Calibration

Preferences

- Preferences

$$u(c_t) \; + \; v(T_t) \; = \; \frac{c_t^{1-\sigma}}{1-\sigma} \; + \; \frac{\left(1+T_t^2\right)^{-(1-\sigma)}}{1-\sigma}$$

- Global temperature change (relative to pre-industrial levels).

$$T_t \; = \; \eta\left(\psi ln\left(\frac{Z_t}{\overline{Z}}\right)\right)$$

$\overline{Z}$: Pre-industrial atmospheric carbon concentration $\approx$ 596 Gtn

ψ : Equilibrium climate sensitivity

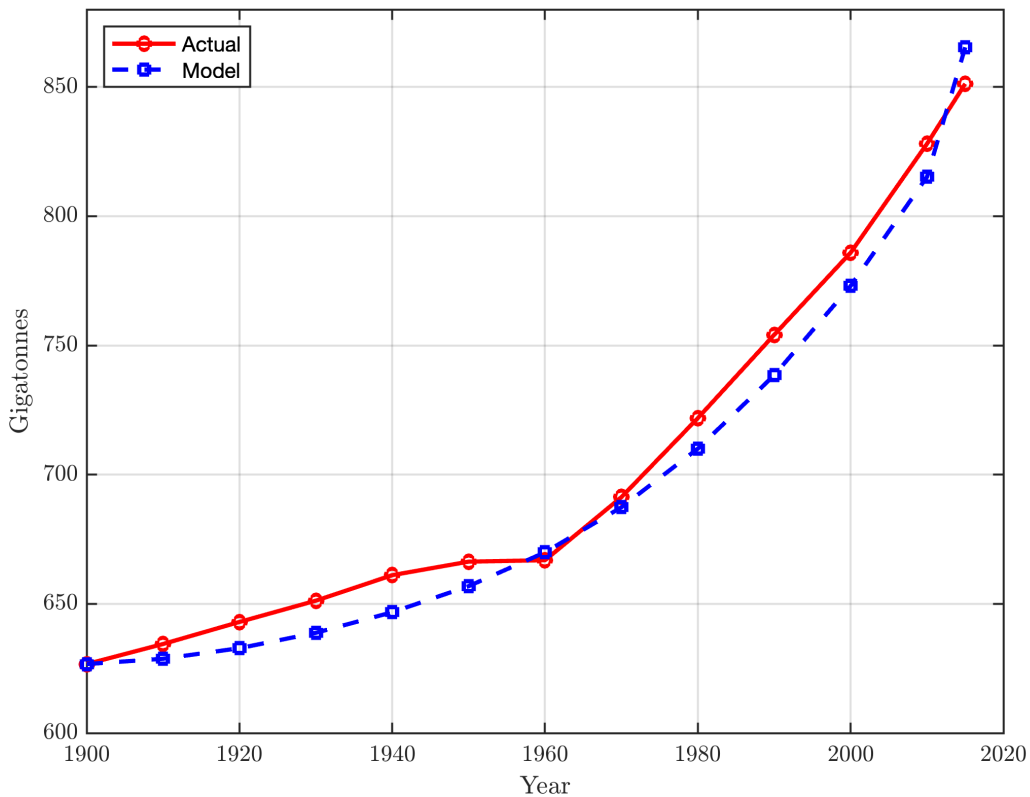
(Change in temperature following a doubling in carbon concentration)

Carbon Cycle

$$Z_t = \sum_{l=0}^{t} (1-\delta_l)(K_{t-l} + K^*_{t-l})$$

$$1-\delta_l = \varphi_p + (1-\varphi_p)\varphi_0\varphi^l$$

$\varphi_p = 0.2$: 20% of emission stays in the atmosphere permanently.
$\varphi_0 = 0.794$; $\varphi = 0.758$ to match historical carbon concentration.



Price of Exhaustible Resource

$$p_t^e = \left(\prod_{s=1}^{t} (1 + r_s^w) \right) p_0^e$$

Using historical oil price data between 1900 and 2005 to compute annual price increase r^w.

The 10-year average price increase is 43%, giving $r^w = 3.6\%$

Initial Technology Gaps

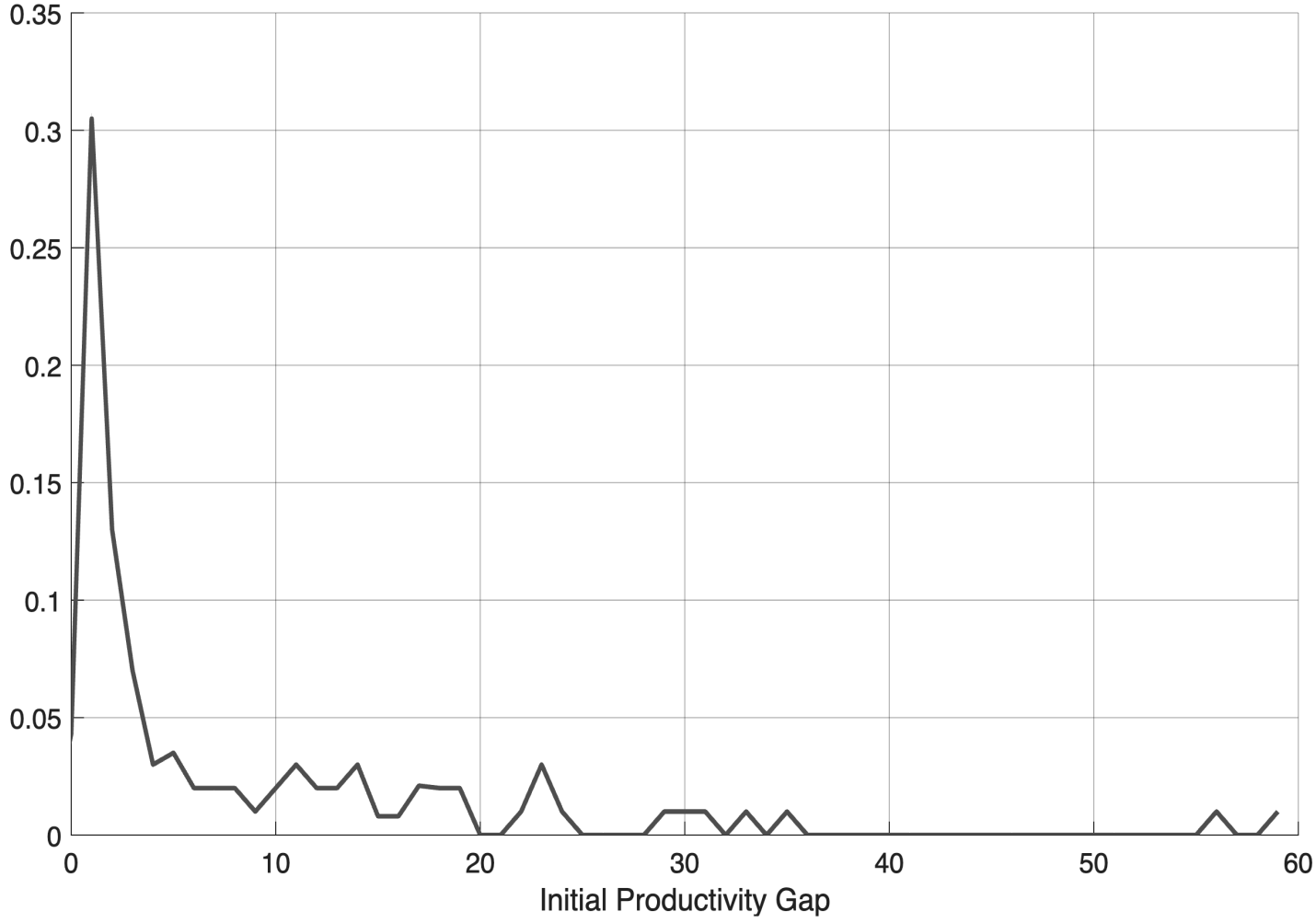
Distribution of initial gaps from Acemoglu et al. (2012)

- Patents are classified as "clean" or "dirty" using data from the U.S. energy sector (1975-2004).

Clean-energy patents	Dirty-energy patents
Geothermal, Hydroelectric	Extraction, Refinement
Nuclear, Solar & Wind energy	Use of Fossil Fuel–based Energy
Conservation & Energy Utilisation	(Oil, Coal, Natural gas & Shale technologies)

- Sectors with a lot dirty patents are assigned a larger initial gap, while sectors with more clean patents have a smaller gap.

Initial Technology Gaps



Parameters

Parameter	Description	Value	Source or Target
σ	CRRA coefficient	1.5	Nordhaus (2008)
ψ	Temp change sensitivity	4.33	ECS $= 3°$C
L^s	Measure of scientists	0.055	Acemoglu (2016)
β	Discount rate (HH)	$(0.985)^{10}$	Nordhaus (2008)
ρ	Discount rate (Firms)	$(0.985)^{10}$	Nordhaus (2008)
γ	Exhaustible share	0.04	Golosov et al. (2014
r^w	World interest rate	0.036	Historical oil prices
φ_p	Permanent carbon %	0.2	IPCC (2007)
φ_0	Share of non-permanent emission decaying	0.794	Historical carbon ac
φ	Rate of decay of non-permanent emission	0.758	Historical carbon ac
α	Breakthrough prob.	0.04	Acemoglu (2016)